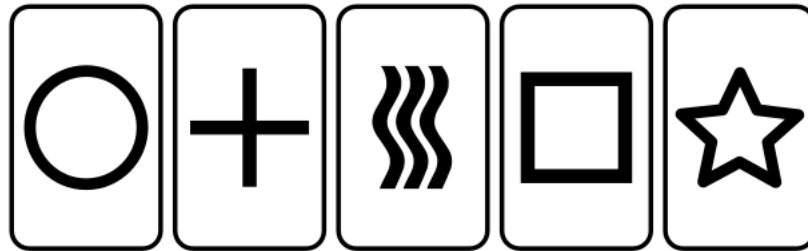


Quiz 13

Some people believe in clairvoyance, a type of psychic power or extra-sensory perception (ESP). A common way for believers to test for ESP is with Zener cards (picture from Wikipedia):



These cards have five “suits,” and someone being tested for ESP tries to guess the suit without seeing the card.

The Amazing Randi (look him up!) tested Valerie Swan, a crystal healer who claimed to have ESP. He had guess the suits of 250 randomly (and independently) chosen Zener cards.

1. Randi was an ESP skeptic. What probability distribution would he use to model the number of cards that Swan would get right?

He would use the distribution $\text{binomial}(250, 1/5)$ for the random variable N representing the number of cards that Swan gets right.

2. I don’t know how many cards Swan got right. Imagine she got 58 cards. Calculate a p-value using Randi’s probability model. Show your work! What conclusion do you draw about Swan’s psychic powers?

The expected number of cards for Swan to get according to Randi’s probability model is $250(1/5) = 50$. The actual number of cards is thus in the right tail. To test the model, we calculate the tail probability (p-value) $P(N \geq 58)$.

Given Randi’s probability model for N , we have $E[N] = 250(1/5) = 50$, $\text{Var}(N) = 250(1/5)(4/5) = 40$, and $\sigma(N) \approx 6.32$.

Since N is the sum of 250 iid random variables (indicators for “guess i is correct”), it is reasonable to treat N as approximately normally distributed. We use the normal approximation with histogram correction to estimate $P(N \geq 58)$:

$$\begin{aligned} P(N \geq 58) &= P(N \geq 57.5) \approx P(\text{standard normal} \geq 1.19) = 1 - P(\text{standard normal} \leq 1.19) \\ &\approx 1 - 0.8830 = 0.1170. \end{aligned}$$

Under Randi’s model, the probability that Swan would get at least 58 suits right is about 12%. The conventional cutoff for worrying about a model is p-value 0.05 or 5%. Since the p-value is greater than 5%, Swan’s performance would not make Randi reject his skeptical probability model. We do not have convincing evidence for Swan’s psychic powers.